MIL-M-48184A(AR)
4 NOVEMBER 1977
SUPERSEDING
MIL-M-48184(PA)
30 March 1976

MILITARY SPECIFICATION

MUNITION FOR PROJECTILE, 155MM, M692/M731;

LEAD CUP ASSEMBLY FOR

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, loading, assembling and packing for one type of Lead Cup Assembly used in the munition for Projectiles, 155MM, M692 and M731.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-650 - Explosive: Sampling, Inspection and Testing

DRAWINGS (SEE 6.5)

*U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9245691	- Detonator, Electric, M100
9298456	- Lead Cup Assembly
9298406	- Detonating Cord Assembly
9298464	- S&A Assembly
7548103	- Box, Set-up, Packing, Ammunition, Outer for Small Explosive Components w/o Primary Explosives
7548104	- Box, Set-up, Packing, Ammunition, Inner for Small Explosive Components w/o Primary Explosives
9321552	- Packing and Marking for Box, Packing

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-190 (latest revisions)").

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Lead Cup Assembly.-The lead cup assembly shall comply with all requirements specified on drawing (dwg) 9298456, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Lead cup assembly output.-The lead cup assembly shall produce a 0.015 inch minimum (min) depression in a 6061-T6 aluminum plate with 1/4 inch minimum thickness.

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3.4 Lead cup assembly functioning.-The lead cup assembly shall function the detonating cord assembly high order as evidenced by detonation of the detonating cord for a minimum of four (4) inches.

3.5 Temperature conditioning and functioning of lead cup assembly.-

3.5.1 Hot temperature.-The lead cup assembly shall function the detonating cord assembly high order as specified in 3.4 after conditioning at 165 + 5°F and while at a conditioned temperature of 125 ± 5°F.

3.5.2 Cold temperature.-The lead cup assembly shall function the detonating cord assembly high order as specified in 3.4 after conditioning at -25 ± 5°F and while at the conditioned temperature.

3.6 Waterproofness test of lead cup assembly.-The lead cup assembly shall be waterproof as evidenced by proper output after being immersed in water maintained at 70 ± 10 degrees Fahrenheit(°F) at a depth of 2 inches minimum from the surface to the top of the assembly for a period of 40 hours minimum.

3.7 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter that would impair proper functioning. Cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Disc, Closure	9298457	25
Cup, Lead	9298458	25
Lead Cup Assembly	9298456	175 (Note)

Note: Fifteen (15) of these assemblies shall be assembled using the procedure specified in 4.5.3 to obtain the density of the explosive charge.

4.3.2 Inspections to be performed - See MIL-A-48078 and Table 1 herein.

*4.3.3 Rejection.-See MIL-A-48078 and Table 1 herein.

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*TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Lead Cup Assembly and Components	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc, Closure (Dwg. 9298457)				
	Examination for Defects	25	-	3.2	4.4.2.1
	Cup, Lead (Dwg. 9298458)				
	Examination for Defects	25	0	3.2	4.4.2.2
	Lead Cup Assembly (Dwg. 9298456)				
	Examination for Defects	160	-	3.2	4.4.2.3
	Functioning Output	50	50-0-1	3.3	4.5.4
	Functioning (Ambient)	20		3.4	4.5.5.4
	Hot temperature conditioning and functioning	20	60-0-1	3.5.1	4.5.5.2
	Cold temperature conditioning and functioning	20		3.5.2	4.5.5.3
	Waterproofness	50	50-1-2	3.6	4.5.6
	Density of explosive charge	15	15-0-1	3.2	4.5.3

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4.4 Quality Conformance Inspection.-

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of lead cup assemblies shall contain:

a. PBXN-5, seived in accordance with Dwg. 9298456, from one batch or blending of batches which results in homogeneity of PBXN-5 used in loading the entire lead cup assembly.
b. Parts of one lot interfix number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48184A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1 *	Disc, Closure	1 OF 1	9298457
			NEXT HIGHER ASSEMBLY 9298456

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Thickness		0.40%	3.2	Gage
Minor 201	Diameter, min. (See Note 1)		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

1. See Paragraph 4.5.1.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48184A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2 *	Cup, Lead		9298458
			NEXT HIGHER ASSEMBLY 9298456

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	(See 4.5.1)			
Major					
101	Thickness of bottom		0.40%	3.2	Gage
102	True position of inside diameter with outside diameter		0.40%	3.2	Gage
103	Inside diameter		0.40%	3.2	Gage
Minor					
201	Inside finish improper		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48184A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Lead Cup Assembly		9298456
			NEXT HIGHER ASSEMBLY

CATEGORY		EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION-METHOD
Critical		None defined				
Major						
	101	Explosive on exterior surface		100%	3.2	Visual
	102	Closure disc punctured		100%	3.2	Visual
	103	Cup split or cracked		100%	3.2	Visual
	104	Functioning output	50	50-0-1	3.3	4.5.4
	105	Functioning (ambient)	25		3.4	4.5.5.5
	106	High temperature conditioning and functioning	50	125-1-2	3.5.1	4.5.5.2
	107	Low temperature conditioning and functioning	50		3.5.2	4.5.5.3
	108	Density of explosive charge			3.2	4.5.3
	109	Moisture and volatiles content of explosive charge			3.2	4.5.2
	110	Total length		0.40%	3.2	Gage
	111	Outside diameter		0.40%	3.2	Gage
	112	Lacquer seal missing, incomplete or on wrong end		0.40%	3.2	Visual
Minor						
	201	Crimp less than 360°		0.65%	3.2	Visual
	202	Assembly damaged		0.65%	3.2	Visual
	203	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
* 4.4.2.4	Box Set-up, Inner, prior to sealing	1 OF 1	7548104
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of lead cup assemblies		0.40%	3.2	Visual
102	Cushion missing or inverted		0.40%	3.2	Visual
Minor					
201	Fillers missing		0.65%	3.2	Visual
202	Marking incorrect or unidentifiable		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
* 4.4.2.5	Box, Set-up, Outer, prior to sealing	1 OF 1	7548103
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of inner boxes		0.40%	3.2	Visual
102	Fillers missing		0.40%	3.2	Visual
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
* 4.4.2.6	Box, Set-up, Outer, Sealed prior to inserting into bag	1 OF 1	7548103
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major:					
101	Corners not blunted		0.40%	3.2	Visual
102	Sealing strip missing		0.40%	3.2	Visual
Minor					
201	Marking incorrect or unidentifiable		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
* 4.4.2.7	Sealed barrier bag	1 OF 1	7548103
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Heat seal of barrier bag missing or incomplete		0.40%	3.2	Visual
102	Barrier bag torn or punctured		0.40%	3.2	Visual
Minor					
201	Marking incorrect or unidentifiable		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
* 4.4.2.8	Packing and Marking for Box, Packing, prior to sealing		9321552
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of bagged outer boxes		0.40%	3.2	Visual
Minor					
201	Fillers missing		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48184A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
* 4.4.2.9	Box, Packing, Sealed		9321552
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged exposing contents		0.40%	3.2	Visual
102	Strapping missing, broken or loose		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Visual-Manual
202	Marking incorrect or unidentifiable		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See MIL-A-48078.

4.5 Test Methods and Procedures.-

4.5.1 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Moisture and volatiles content of explosive charge.-The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture and volatiles content of the explosive charge shall be determined at the loading station at the time of loading. This determination shall be made at least once each four (4) hours of production. Sampling and testing of the explosive charge shall be in accordance with MIL-STD-650, Method 101.5. In addition, if the moisture and volatiles content fails to comply with the requirement specified on the applicable drawing and loading has not begun, the explosive charge material shall be rejected until corrective action has been taken as verified by repeating the moisture and volatiles content determination If loading has begun, lead cup assemblies loaded with the non-conforming explosive charge shall be rejected.

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4.5.3 Density of Explosive Charge.-A minimum of two (2) assemblies, representative of the lead cup assemblies being processed, shall be loaded and the densities determined to qualify each press prior to its use in production. In addition, a sample of two (2) assemblies shall be processed during the last portion of each one (1) hour's production from each press and the density determined as a control of each press during production.

If any defective assembly is found when inspection is performed for control of a press, the press producing the defective assembly shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All presses removed from production because of some fault, may, after corrective action, be returned to production providing they pass the qualification test above.

The following procedure shall be used to determine the density of the explosive charge.

a. Weight empty lead cup (grams)

b. Measure bottom thickness (inches)

c. Measure wall thickness (inches)

d. Load cup

e. Weight loaded lead cup (grams)

f. Measure height to top of charge (inches)

g. Measure outside diameter of loaded lead cup (inches)

h. Calculate weight of charge (e-a) (grams)

i. Calculate height of charge (f-b) (inches)

j. Calculate volume of explosive charge (cubic centimeters) = (.7854) x (16.39) x (g-2c) x (g-2c) x (i)

k. Calculate density of explosive charge (above). (grams per cubic centimeter) = (h/j)

If any assembly fails to comply with the density requirements specified on the applicable drawing, it shall be classed defective. This is a destructive test. Assemblies so tested shall not be returned to the lot.

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NOTE: When weighing in water, use distilled water containing a suitable wetting agent, e.g. "Aerosol" (1cc per 250cc distilled water)

4.5.4 Functioning output.-The lead cup assemblies shall be assembled into a plastic holder and placed with the green end of the assembly in contact with a witness plate of material and thickness as specified in 3.3. The lead cup assembly shall be initiated by an M100 electric detonator (Dwg. 9245691) firing across an air gap of .230 ± .010 inches. The detonator shall be initiated by applying the energy from a suitable power source (e.g. 100 microfarad capacitor charged to 1.6 volts maximum or 6 volt battery). The depth of the depression in the witness plate shall be determined using a dial indicator with a .015 ± .001 inch spherical radius indicator tip.

4.5.5 Conditioning and functioning.-

4.5.5.1 Preparation of samples.-Each lead cup assembly to be subjected to the functioning test shall be assembled into the output hole of an S&A assembly (Dwg. 9298464 without slides, slide springs, barrier assemblies, setback pin and spring) and staked in place with the seal washer. An electric detonator M100 (Dwg. 9245691) shall be shorted and assembled with seal washers to the input hole of the S&A assembly.

4.5.5.2 Hot conditioning.-The lead cup assemblies to be conditioned and functioned at hot temperature shall be prepared as specified in 4.5.5.1. They shall then be placed in a suitable temperature conditioning box and the temperature increased to 165 + 5°F with relative humidity not less than 15%. The rate of temperature increase shall not exceed 18°F per minute. The samples shall be maintained at temperature for forty (40) hours minimum.

The temperature in the oven shall then be reduced to 125 ± 5°F and the assemblies shall be conditioned at temperature for two (2) hours minimum. The assemblies shall then be removed from the oven and functioned as specified in 4.5.5.4. Set-up time between removal from the oven and firing shall be five (5) minutes maximum.

4.5.5.3 Cold conditioning.-The lead cup assemblies to be conditioned and functioned at cold temperature shall be prepared as specified in 4.5.5.1. They shall then be placed in a suitable chamber and the temperature decreased to -25 ± 5°F. The rate of

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decrease shall not exceed 18°F per minute. The assemblies shall be conditioned at temperature for two (2) hours minimum. The assemblies shall then be removed from the chamber and functioned as specified in 4.5.5.4. Set-up time between removal from the chamber and firing shall be five (5) minutes maximum.

4.5.5.4 Functioning.-Each assembly prepared as specified in 4.5.5.1 and after applicable temperature conditioning shall have a detonating cord assembly (Dwg. 9298406) from an accepted lot assembled to the output side of the lead cup assembly with the face of the booster on the detonating cord assembly axially aligned and parallel to the face of the lead cup assembly. It is not necessary to temperature condition the detonating cord assembly. The space between the assemblies shall be .020 inches minimum. The detonating cord shall be taped to an aluminum plate (approximate dimensions .062" X 2" X 4 1/2"). The detonator shall be functioned behind an approved barricade. The detonator short shall be removed and the detonator shall be initiated by a suitable power source (e.g. 6-volt battery). If the lead cup assembly fails to initiate the detonating cord or fails to cause high order detonation of the detonating cord for the minimum length specified in 3.4, as evidenced by a witness mark on the aluminum plate, the lead cup assembly shall be classed defective. Any assembly in which the detonator fails to fire shall be classed as a "no test" and a replacement S&A assembly as specified in 4.5.5.1 shall be used in its place.

4.5.6 Waterproofness test. - The lead cup assemblies shall be immersed in water at a depth of 2 inches minimum from the surface of the water to the top of the assemblies. The water temeprature shall be maintained at 70° ± 10°F. After a minimum of 40 hours, the assemblies shall be removed from the water and subjected to the output test specified in 4.5.4. The functioning test of all samples shall be completed within four hours after the assemblies are removed from the water.

WARNING: The above tests are conducted on assemblies containing an explosive component. Test barricades (where used), procedures and equipment shall have prior approval of the contractor's safety officer.

5. PACKAGING

5.1 Preservation and Packaging.

5.1.1 Level C. - The lead cup assembly shall be preserved and packaged in accordance with Dwg. 7548104.

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5.2 Packing. -

*5.2.1 Level C. - The lead cup assembly, preserved and packaged as specified in 5.1.1, shall be packed in accordance with Dwgs. 7548103 and 9321552.

*5.3 Marking. - Marking shall be in accordance with Dwgs. 7548103, 7548104, 9321552 and Code of Federal Regulations, Title 49, Parts 100-199.

5.4 Shipping.-When components from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use in the Munition for the M692 and M731 Projectiles.

6.2 Ordering data.-See MIL-A-48078.

*6.3 Submission of inspection equipment design for approval. - See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRDAR-QAA_I, ARRADCOM, Dover, New Jersey 07801. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Check test for possible deterioration.-The following test is designed to prevent lead cup assemblies that have deteriorated from being assembled into major ammunition items. The test shall be performed on lead cup assemblies for each lot which have been stored for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the lead cup assemblies are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

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*6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data orignially prepared by these activities is now under the cognizance of ARRADCOM.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (addition, modifications, corrections, deletions) from the previous specification were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespictive of the marginal notations and relationship to the previous specification.

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